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Activities like sharing of thoughts, advertising business, connecting with peers, and staying updated in this world are highly facilitated by social media platforms. In the unique form of media called memes, information is conveyed through the image-to-text or text-to-image dependency relationship. Popular memes are often driven by viewers rather than marketing or advertising techniques, which indicates the level of engagement of social media users with memes. Given the popularity of memes, a demand for a solution to identify and counteract hate-spreading memes on social media platforms is raised. In this study, a multimodal machine learning approach to detect offensive memes is presented, where the text of memes are spelled in the Devanagari script of the Hindi language. A dataset of 9262 images has been created, and they have been labeled as offensive or not offensive. As the dataset is highly imbalanced, another dataset which has 3732 images is created by undersampling the total dataset and the models are trained for both the imbalanced dataset as well as the balanced dataset. Finally, the classification problem is solved through a multimodal Logistic Regression classifier that utilizes concatenated feature representations of image and text. An accuracy of 0.81 is achieved by the model.

Additional Key Words and Phrases: Multimodal, Hindi memes dataset, Offensive memes, Classification

1 Introduction

Memes are one of the popular forms of media that are widely shared on social media platforms like Instagram, Facebook, Twitter, etc., and are loved mostly by children and youth. They are unique in the sense that information through memes is conveyed via image-to-text or text-to-image dependency relationship. Memes can be in the form of images, GIFs, or videos. Meme content is majorly based on the day-to-day activities of people, daily political or socio-economic updates of any country or the world, etc. Memes generally contain humor or sarcasm, but in their own unique way, they may prove offensive by targeting a specific community or a person based on their religion, caste, or any other specific features.

1.1 Popularity and Impact of Memes

According to an article by Gurus Coach, memes generate ten times more engagement than any other form of media, and also, 55% of 13-35-year-old social media users send memes every week. 73% of smartphone users share memes on social media platforms. The article also suggests that meme is the second most likely shared form of media on

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these platforms [8]. The hate speech targeted towards a community can lead to online harassment and can be a major source of cyberbullying. In a survey by ADL, more than one-fourth of Americans have experienced severe online harassment [1] and about 44% have been a victim of cyberbullying. There are no existing government or internet laws to check upon hateful memes; however, some social media platforms like Facebook and Twitter have enforced rules regarding the conduct of hateful speech on these platforms in a generic approach to check offensive posts and comments [26]. Figure 1 is a typical example of a Meme. The image of the meme shows a picture of a couple and the meme text spelled in the Hindi language translates nearly to "A grasshopper destroying the healthy crops", signifying the racist approach to the meme image.

Classification of a meme as offensive or benign is a multimodal problem statement since the complete information of the meme is conveyed through both image and text. As in Figure 1, the information shared by text and image of the meme, when taken into consideration separately, may not be offensive. However, when the information conveyed through both of them is analyzed along with the knowledge of racism, the information conveyed by the meme as a whole becomes clear.

Fig. 1. A dataset sample

1.2 Popularity and Impact of Hindi Memes in India

Hindi is one of the official languages of India and there are around 400 million Hindi-speaking people in India which is 35% of the Indian population [40]. Social media users in India stood at 518 million in India in 2020 and are expected to rise to 1.5 billion by 2040 according to the statistics from Statista [34]. Memes, being a popular form of media on these platforms, a dire need arises to identify and counteract the offensive memes so as to make social media platforms a healthier space for interaction. The popularity of memes in India has given rise to a whole new set of content and community which is now called "Meme Culture". This "Meme Culture" in India has given rise and popularity to an all new set of content creators, meme characters and also the development of various meme-creating apps that allows the creators to either create a meme from scratch or to add their idea to an existing meme template. Several studies and articles suggest that memes are so popular and powerful that they can create a great impact in the minds of viewers to the extent that they can be held accountable for change in the perception of people in regard to various social, political, and religious issues. The two popular sources of entertainment in India- Cricket and Bollywood also provide a good amount of subjects to meme creators. However, in recent times, it has also been noted that content creators tend to create content based on false news, unverified data, and rumors in order to increase engagement on their page and gain more popularity [19]. The detrimental usage of memes on sensitive issues can cause havoc due to the massive popularity of this form of media.

1.3 Research Questions

The research work aims to enhance our understanding in the domain of multi modal classification of memes, simultaneously taking into account the text of yet another low-resource but popular language, Hindi.
RQ 1: Attribution of the nature of memes as multi-modal. The underlying message conveyed by memes can be understood as a concatenation of information conveyed by the image and the information conveyed by the text, at the same time holding a pre-requisite knowledge about the context of existing nature of society, politics and culture, etc. The text and image features may or may not contribute equally towards the final information conveyed by a meme. Sundarajan *et al.* proposed Integrated Gradients method to score the contribution of independent input features on the final predicted results [35]. A number of studies that are based on a similar problem statement take into account the visual and linguistic features of the meme and attempt to perform the classification task [32][14]. In this research work, we have applied both unimodal and multimodal approaches, only to arrive at the conclusion that the multimodal approach produce better results.
RQ 2: Existence of slur language in text. To identify offensive and hate-spreading text in the form of comments, post-captions or meme text on social media platforms is notoriously challenging due to the added slur language present in the text. Slur words can be regional as well. To identify the existence of derogatory or hateful slur words, a vast background knowledge is required. Existing studies based on identifying hateful Hindi text undertake it as an NLP task to classify given text as offensive or benign. Velankar *et al.* achieve a significant accuracy in Hindi text classification task by utilising models like indicBERT, LSTM among others [37]. We utilise models like LASER, LaBSE and multilingual BERT to classify text as offensive or otherwise.

1.4 Contributions

The major contribution of this paper can be summarised as follows:
1) Create a labeled dataset of popular memes where the text of the meme is spelled in Hindi from different social media platforms.
2) Extract the text from the meme images and classify memes as offensive or benign using a unimodal approach for text classification.
3) Classify the memes as offensive or benign using a unimodal approach for image classification.
4) Use a multimodal approach to classify memes as offensive or benign by using concatenated feature embeddings of images and text.

Although there are a number of studies focusing on the fusion or concatenation of visual and lingual features, but the problem statement here not only requires the concatenation of the features of image and text but also significant reasoning to relate them. To discover the implicit relationship between the two is still a challenging task. To address and counteract this challenge, we have proposed the multimodal classifier that trains the model based on concatenated embeddings of image and textual features.

1.5 Novelty

The Novelty of this work is:
1) There is no publically available labeled dataset based on this problem statement of detecting offensive content in Hindi memes. A dataset of 9262 images is created and labeled based on fixed annotation criteria.
2) Separate unimodal approaches on meme-text and meme-image are applied using pre-trained models to classify offensive and benign memes.
3) A multimodal classifier is trained based on visual and textual embeddings of the meme.

1.6 Organization of Research Paper

The organization of the research paper is as follows -

- Related Work - This section contains the previous work on different datasets related to the detection of offensive content in memes.
- Dataset - The dataset section focuses on the collection, annotation, and pre-processing of data.
- Methodology - The methodology section showcases various procedures followed to get the results. It involves the methods used for text extraction and discusses the models used in detail.

- Results and discussion - This section showcases the results and discusses the performance metrics used to evaluate the problem.
- Conclusion and Future Work - The last section talks about the overall results obtained during the course of this study and future scope of the project.

2 Related Work

Detection and classification of offensive memes have gained significant attention pertaining to competitions like the Facebook Hateful Meme Classification Challenge and Dravidian LangTech 2021. Table 1 provides a summary of literature review.

2.1 Hateful Memes Classification for English Memes

Sandulescu *et al.* fine-tuned the pre-trained VL-BERT, VLP and UNITER and dual stream models such as LXMERT to improve accuracy [29]. Shang *et al.* developed a knowledge-enriched graph neural network solution to effectively counter multimodal memes detection problem [31]. Giri *et al.* designed a deep learning-based NLP model to detect offensive memes in three steps: extract text from the image, classify it as offensive or not, if offensive- finally label it as slightly, very or hateful offensive [14]. Badour *et al.* approached the problem by converting both textual and visual channels to a combined vector representation and training them using the bi-directional LSTM model in the first approach. In the second approach, both the vector representations were fed into SentenceBERT to get an embedding which was later classified citebadour2021hateful. Deshpande *et al.* in their multimodal approach, focused on feature selection to build a gradient-boosted decision tree and an LSTM-based model [11]. Aggarwal *et al.* utilized VL-BERT to classify hateful memes [2] and Wanbo et al. used pre-trained VisualBERT and ViLBERT models and extended further to the usage of MSRD method to improve accuracy [39]. Chen *et al.* developed the OSCAR+RF model to conclude that Vision-Language PTMs with the addition of anchor points can improve the performance of deep learning-based detection [7]. Blaier *et al.* demonstrated a caption-enriched approach to improve the accuracy of pre-trained models. [6] Mathias *et al.* divided the task into two subtasks, namely identifying the protected category attacked by meme and identifying the attack type for fine-grained detection of the target [25]. Arya *et al.* utilised CLIP model to analyze the text and images, and to draw conclusion if combined expression qualifies as hateful [3].

2.2 Hateful Memes Classification for non-English Memes

A few substantial efforts to identify hateful content in memes in which meme text is not in English and/or meme language is not English have been done in recent times. Kumari *et al.* worked on offensive hindi memes detection problem using MMCLIP model and achieved an accuracy of 83.71% [21]. Silva *et al.* developed an automated tool to detect hate content in images containing Sinhala (an official language of Sri Lanka) texts. The images were not necessarily memes [33]. Maity *et al.* created Hinglish (Hindi written in English) memes dataset and proposed two different multimodal frameworks: BERT+ResNET-Feedback and CLIP-CentralNet to detect hate speech leading to cyberbullying [24]. Hossain *et al.* employed weighted ensemble techniques to assign weights to the participating visual, textual, and multimodal models like VGG19, VGG16, multilingualBERT, etc. on multiOFF [36] and TamilMemes dataset [16]. Ghanghor *et al.* used customized versions of transformers-based pre-trained models on datasets containing Tamil, Malayalam, and Kannada memes [13]. Hossain *et al.* proposed multimodal deep neural network based model DORA (Dual cO-attention fRAmework), to extract significant modal features from Bengali memes and evaluate them [17].

2.3 Research Gap

To the best of our knowledge, there has been only a single published study that focused on the Hindi meme classification problem. Till the date of submission of this paper, there is no publicly available dataset of Hindi memes. A labelled dataset was created for this research work. None of the studies reported the result of K-Fold Cross Validation using multimodal approach.

Citation	Major Approaches	Dataset	Best Result
[21]	MMEM+MFB and MMCLIP models	7416 Hindi memes	Accuracy 0.83
[31]	Knowledge-enriched graph neural network	Memes dataset from Reddit and Gab	Accuracy 0.73
[14]	Deep learning based NLP model	6992 memes from different social media platforms	Accuracy 0.80
[18]	FixEfficientNet-L2 model	9540 memes from Facebook AI team	Accuracy 0.63
[41]	Triplet-relation network to model relationship between image and text	Facebook Hateful Meme Dataset	AUROC 0.78
[4]	Bi-directional LSTM model and SBERT	Facebook Hateful Meme Dataset	Accuracy 0.75
[33]	NLP-CNN	2000 data points using Facebook APIs	Accuracy 0.72
[10]	Bi-directional LSTM and CNN	Facebook Hateful Meme Dataset	Macro F1 0.249
[32]	Analogy-aware Offensive Meme Detection (AOMD) framework	Memes dataset from Reddit and Gab	Accuracy 0.71
[11]	Gradient-boosted decision tree and LSTM-based model	Facebook Hateful Meme Dataset	Accuracy 0.71
[24]	Two different multimodals: BERT+ResNET-Feedback and CLIP-CentralNet	25000 Hinglish memes from Facebook, Twitter, Reddit	Accuracy 0.74
[43]	VL transformer architecture ensemble: VL-BERT,UNITER,VILLA and ERNIE-Vil	Facebook Hateful Meme Dataset	AUROC 0.77
[27]	Vilio: a visual-linguistic model	Facebook Hateful Meme Dataset	AUROC 0.82
[38]	VisualBERT with Ensemble Learning	Facebook Hateful Meme Dataset	AUROC 0.81
[23]	Three different models: LXMERT, UNITER and Oscar	Facebook Hateful Meme Dataset	AUROC 0.80
[29]	Three different models: Fine tuned VL-BERT, VLP, UNITER and LXMERT	Facebook Hateful Meme Dataset	AUROC 0.79
[28]	BERT with Multi-layer Perceptron	5020 English memes from Google images	Accuracy 0.83
[22]	DisMultihate multimodal	Facebook Hateful Meme Dataset and MultiOFF	AUROC 0.82
[5]	VGG-16, Inception-V3, LSTM, BERT classifier based model	6,992 and 1,878 memes train and test data	F1 score 0.33
[16]	Weighted ensemble technique and fusion approach to develop a multimodal model	MultiOFF and TamilMemes	F1 score 0.65
[6]	Caption-enriched approach to improve accuracy of BERT, ViLBERT, VisualBERT	Facebook Hateful Meme Dataset	AUROC 0.78
[7]	OSCAR+RF	Facebook Hateful Meme Dataset	AUROC 0.77
[42]	Transformer-based multimodal with data augmentation, ensemble learning	Combination of memes based on COVID-19 and US Politics	Macro F1 54.7
[13]	Customised transformers-based pre-trained models	2300 total memes of Tamil, Malayalam and Kannada	Weighted avg F1 0.55
[3]	Multimodal Hate Speech Detection in Memes Using Contrastive Language-Image Pre-Training	Facebook Hateful Meme Dataset	AUROC 0.88
[20]	Hateful meme classification of Indian political memes	Bharatiya Political Memes Dataset	Accuracy 0.71
[17]	Deciphering Hate: Identifying Hateful Memes and Their Targets	Bengali Hateful Memes Dataset	Weighted F1 0.72

Table 1. Related Work on Hateful Memes Identification

3 Dataset

For this research, publicly available memes from various social media platforms were used and annotated. Various categories discussed in the coming section were defined and if the meme falls under any category, it was labeled as offensive. Figure 2 shows a benign meme as neither the text nor the image is offensive, whereas Figure 3 has offensive text; hence, it is labeled as an offensive meme.[1]

Fig. 2. Benign

Fig. 3. Offensive

3.1 Data collection

9262 memes were collected from social media platforms Instagram, Facebook, and ShareChat. These social media platforms were selected on the basis of their popularity in India. Popular hashtags were used to find relevant meme pages. For example - #hindimemes #hindijokes #nonveghindi #adulthindijokes #desijokes #desimemes #desitrolls, etc. Memes that had the majority of the text spelled in Hindi were selected while others were discarded.

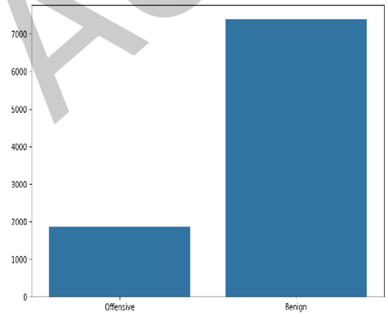

Fig. 4. Dataset distribution

3.2 Data Pre-Processing

After the collection of memes, the text dataset that was obtained from the memes was preprocessed. The dataset had many unwanted characters like @, hash symbol, new lines, punctuation marks, brackets, links, etc. that did not contribute to the results. They were removed before the annotation process.

[1]Dataset

3.3 Annotation

After the pre-processing of the data, the dataset was annotated. Based on references [33] and [25] the criterion was made, falling under which a meme was labeled as "offensive". Incitement of hatred or mockery primarily against a person or a group of people defined in terms of: race, ethnicity, gender, caste, specific mental/health condition/disability, sexual orientation, religious belief, incitement of violence, descent, or specific family background and spreading of vulgar/pornographic content are labelled as offensive. Some memes which were not under any of the given categories but could offend a group of people were removed; for example a political meme. A political meme can be supported by a group of people but can be offensive to another group of people.

3.3.1 Methodology of Annotation. For the purpose of annotation of meme images, we required annotators who understand Hindi to the extent of expertise or nativeness, and are also aware of political, social and cultural environment of India. Three authors of this research work have served as annotators for the dataset. All three annotators are citizens of India since birth and have native proficiency in Hindi language. Annotators have studied the domain of AI for two years and are also avid social-media users. Remaining authors of this research work ensured that there are no cultural, political, religious or gender biases in the labelling task done by the annotators.

3.3.2 Inter-Annotator Agreement. To check the reliability of the work done by the three annotators, we measured the inter-rater agreement by utilising Krippendorff's Alpha coefficient. This statistical metric is best suited for this nominal classification scenario with more than two annotators. It is a general and robust method that indicates how well the measurements done by a given number of people agree to each other and provides the result value in between 0 and 1, 1 being ideally reliable. For the case of three annotators, the sample set should be between 30-60 labels done by each annotator [30]. We attain a score of 0.76 for a given set of 50 sample memes.

3.4 Overview of Data

The dataset used was imbalanced and consisted of 9262 annotated memes, with 7396 as benign(label-0) and 1866 as offensive(label-1). Figure 4 shows the distribution of the total data into benign and offensive memes. The dataset contains images in JPG format and text of all the memes in a CSV file. The text was extracted from the images using Object Character Recognition.

Due to the imbalanced nature of the dataset, it was undersampled, and the models used for imbalanced data were employed for the balanced data. The balanced dataset consisted of 3732 memes, comprising all the 1866 offensive memes and 1866 randomly selected benign memes.

The dataset was divided into train and test in a 3:1 ratio with reference to [33]. The ratio ensures an adequate amount of data for training as well as for testing.

4 Methodology

The methodology can be majorly divided into three parts - Text extraction, unimodals and multimodal.

4.1 Text Extraction

This research project works on the offensive content in memes, and in memes, the text can be offensive, the image can be offensive, or in some cases, both might not be offensive, but combined together, they can become offensive. Hence, the image, as well as the text, plays an important role in determining whether the meme is offensive or not. The dataset was a collection of images with textual data. As explained above, it was important to have text as well as the image to carry out the research work. Hence, text extraction from images and text dataset development was the first step after the collection of memes. EasyOCR is a convolutional neural network(CNN) + bidirectional LSTM(BiLSTM) + Connectionist

Temporal Classification loss(CTC) deep neural network that performs object character recognition (OCR) and is used for text extraction for the project. EasyOCR can work on multiple languages and works well with text on a noisy background. To increase the speed of the text extraction process, multiprocessing was used. Multiprocessing helps execute multiple input values in parallel by fully utilizing the multiple processors available in a machine.

4.2 Unimodal

As discussed, the dataset has two parts- images and text dataset. In this section, we have separately analyzed the unimodal performance of various text and images. These unimodal act as the base for our multimodal approach. Hence, two unimodal were made, one for images and one for text.

4.2.1 Text Unimodal. For the text unimodal, the first step was to create text embedding. Text embedding represents words in the form of real-valued vectors. It is used to find the relation between words. For this research, highly pre-trained models and embedders like LASER and LaBSE [12] embeddings and multilingual BERT were used.

- LASER- It stands for Language-Agnostic SEntence Representations. It was developed by Facebook AI research, and it uses a Bi-LSTM encoder based on an RNN architecture to generate high-quality representations of sentences in multiple languages, including the Hindi language.
- LaBSE - It stands for Language-agnostic BERT Sentence Encoder. It is a model used for text embedding and is based on BERT architecture, a transformer-based neural network model trained on a large corpus of multilingual text. It extends the BERT model architecture and generates sentence embeddings, and like LASER, LaBSE is also used for generating multilingual sentence embeddings.
- Multilingual BERT - It stands for Bidirectional Encoder Representations from Transformers. It is a BERT model which is trained on a large corpus of multilingual texts. By being trained on a diverse and large corpus of multilingual texts, It can encode language-agnostic representations. It can capture cross-lingual similarities and differences during the training and give high-performance results. Figure 6 shows the flowchart of the multilingual BERT model training.

The next stage was to train the model. Various classification algorithms explained below were used to train and test data. For hyperparameter tuning, GridSearchcv was used.

- SVM - Support vector machine is a classification algorithm where dataset points belong to a predefined set of categories. The aim of the algorithm is to find the hyperplane. The learned hyperplane is placed in such a manner that the difference between the two classes with respect to the hyperplane is maximum. The C value, i.e. the penalty parameter range, was set to [0.001, 0.01, 0.1, 1, 10].
- KNN - K nearest neighbor is an algorithm that can be used to make classifications. KNN algorithm classifies the data and categorizes the new data on the basis of the similarity the new data has with the given categories. The parameter n neighbors, i.e. the number of neighbors ranges, was set to [3,5,7,9].
- Logistic Regression - LR uses the sigmoid function. It fits an exponential curve to the data, which enhances the performance of the model.
- Decision Tree Classifier - It is a supervised learning algorithm that can be utilized for multi-class classification. DecisionTreeClassifier class was used for decision tree classification. The maximum depth was set to 800, and the minimum sample split was set to 5.
- Random Forest Classifier - This algorithm creates a set of decision trees from a randomly selected set of the training dataset. Like the decision tree classifier, the maximum depth was set to 800 and the minimum sample split was set to 5.

		Balanced		Imbalanced	
Embedding	**Classification model**	**Accuracy**	**F1-score**	**Accuracy**	**F1-score**
LASER	SVM	0.65	0.65	0.81	0.73
	KNN	0.55	0.51	0.79	0.76
	Logistic Regression	0.66	0.65	0.81	0.76
	Decision Tree Classifier	0.58	0.58	0.70	0.70
	Random Forest Classifier	0.68	0.68	0.81	0.73
LaBSE	SVM	0.69	0.69	0.83	0.80
	KNN	0.65	0.65	0.81	0.78
	Logistic Regression	0.68	0.67	0.82	0.79
	Decision Tree Classifier	0.56	0.56	0.71	0.71
	Random Forest Classifier	0.68	0.68	0.81	0.73
LaBSE+LASER	Logistic Regression	0.68	0.66	0.82	0.77
Multilingual BERT	Multilingual BERT	0.71	0.71	0.83	0.82

Table 2. Text Unimodal Results for balanced dataset

Table 2 shows the performance of various models with LASER, LaBSE, and LASER+LaBSE embedding. Figure 5 shows the pipeline of text unimodal.

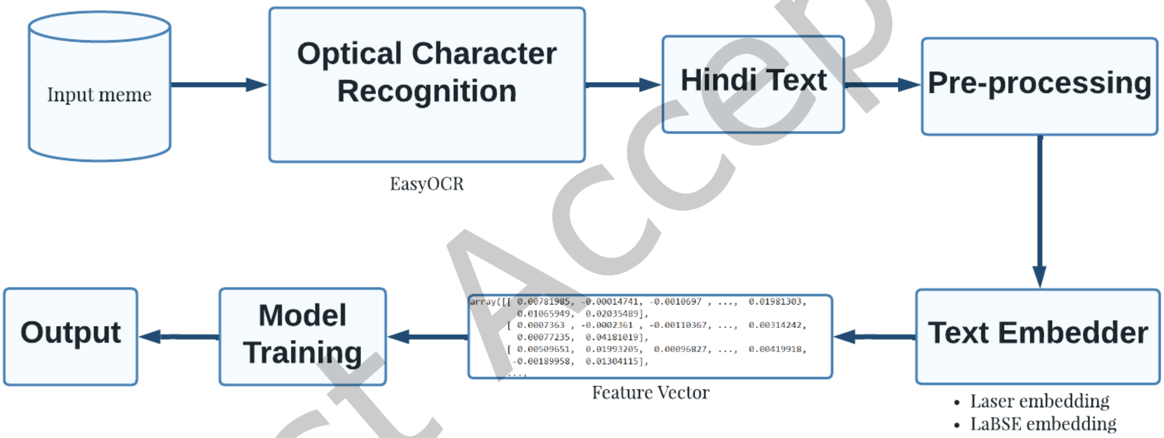

Fig. 5. Unimodal Approach

4.2.2 Image Unimodal. Deep learning neural network-based models, CNN, and VGG16 were used for the image unimodal. These are popularly used for image classification, object detection, etc. Sequential Keras API was used for building CNN. Conv2D, MaxPooling2D, and Dense layers were added to the model, as shown in Figure 7. Here, Conv2D is the convolution layer. In the convolution layer, *Relu* was used as the activation function. MaxPooling was employed to create the pooling layers. Lastly, fully connected layers were created using the Dense function.

VGG16 is a specific CNN architecture consisting of 16 weighted layers, of which 13 are convolutional, and 3 are fully connected layers (Figure 8). VGG16 allows a deeper and more profound representation of features by using small 3x3 convolutional filters throughout the network. Max pooling is used in VGG16, which downsamples the features maps. Because of its deep architecture and homogeneous structure, VGG16 makes it easy to train and fine-tune models for specific tasks.

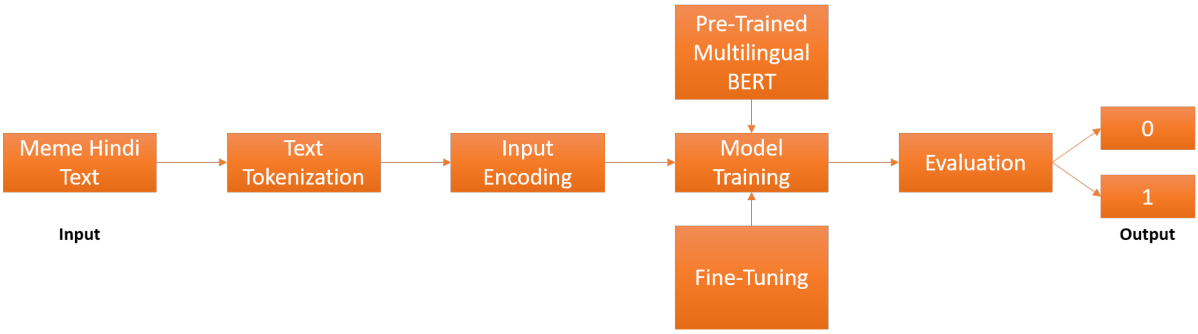

Fig. 6. Multilingual BERT Flowchart

Images were resized and loaded in the array form, then given to our CNN model and VGG16 model as input. Ten epochs were used for training our model for the CNN model and five epochs for the VGG16 model. As a result, for total dataset an accuracy of 84.32% and f1-score of 46.95% with the CNN model and 82.12% accuracy, and 79.65% with the VGG16 model was obtained. For balanced dataset an accuracy of 64.4% and f1-score for 39.75% with the CNN model and 68.92% accuracy, and 68.77% with the VGG16 model was obtained.

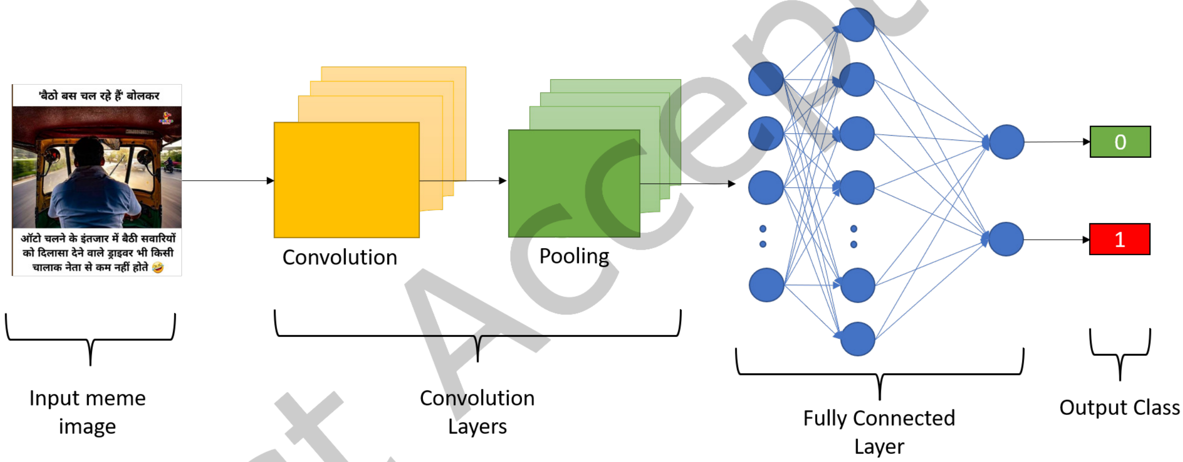

Fig. 7. CNN pipeline for image classification

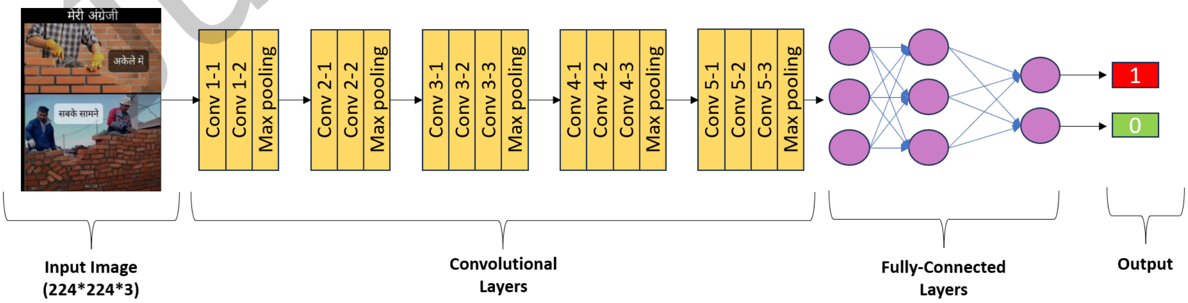

Fig. 8. VGG16 pipeline for image classification

4.3 Proposed work - Multimodal

In the proposed multimodal approach, both images and text present in them were taken into consideration for developing the offensive meme detection model. The text and image unimodal in section 4.2 were analyzed, compared, and used in the proposed multimodal approach by ensembling them. The proposed multimodal approach contains mainly four steps, i.e. text extraction from the image, text, and image embedding, concatenation of both embeddings and model training, as shown in Figure 9. Text extraction is explained in detail in section 4.1. Further steps are discussed in detail in the below subsections.

For the final proposed multimodal approach, i.e., BERT and VGG-16 ensemble model, k-fold validation was used, k being 3. The value of k was chosen based on the size of the dataset. Using 3 folds ensured no overlapping data and an adequate size of the test data. The k-fold cross-validation approach is shown in Figure 10.

For k-fold, the dataset was first divided into training and testing data in a 3:1 ratio such that the training set had 6946 memes and the testing set had 2316 memes. The training dataset was further divided into 3-folds such that all the folds had approximately the same amount of data, and the model was evaluated for all the folds. Lastly, the model was evaluated for the testing data. All the results are tabulated in Table 3.

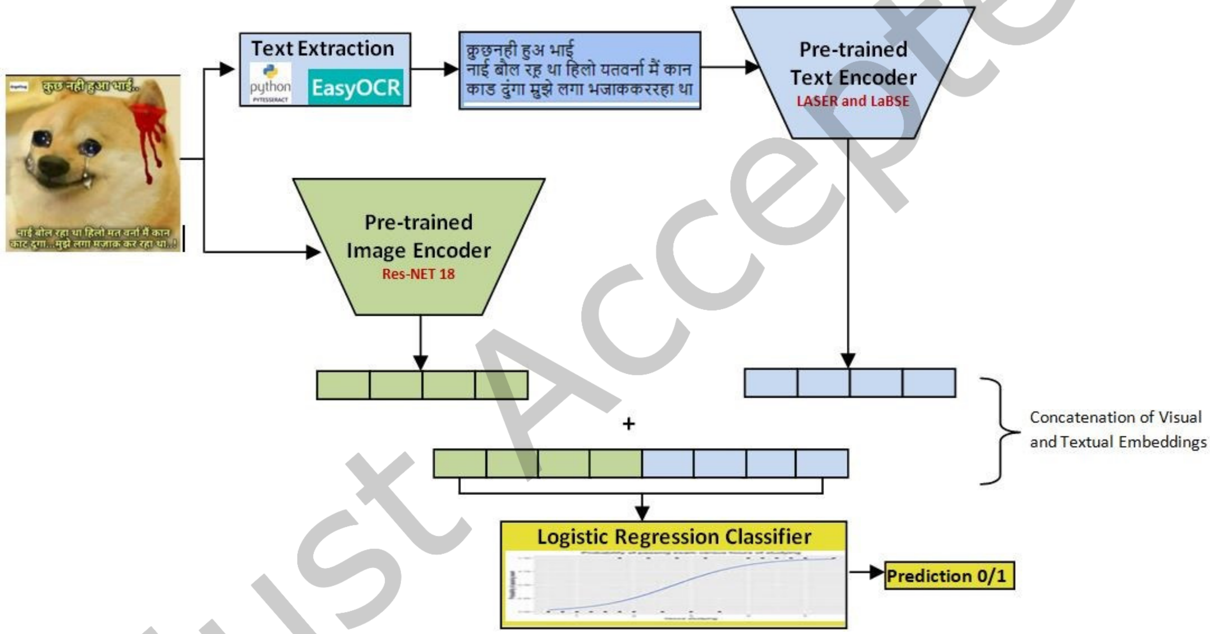

Fig. 9. Multimodal Flowchart

4.3.1 *Text and Image Embeddings.*

For the text embedding, LASER and LaBSE embedders and highly pre-trained transformer-based Multilingual BERT (explained in detail in section 4.2.1) were used [44]. LaBSE(Language Agnostic BERT Sentence Embedding) [12] is a Transformer(BERT) based cross-lingual sentence embedding model which Google AI developed, whereas LASER(Language-Agnostic SEntence Representations) was developed by Facebook AI. LASER and LaBSE support 93 and 109 languages, respectively, including the Hindi language. The Multilingual BERT model is trained on a large corpus of multilingual texts. By being trained on a diverse and large corpus of multilingual texts, it can encode language-agnostic representations and give high-performance results. All three are good for our Hindi text dataset. For image embedding, ResNet-18 and VGG16 were used. ResNet-18 [15] is a type of artificial

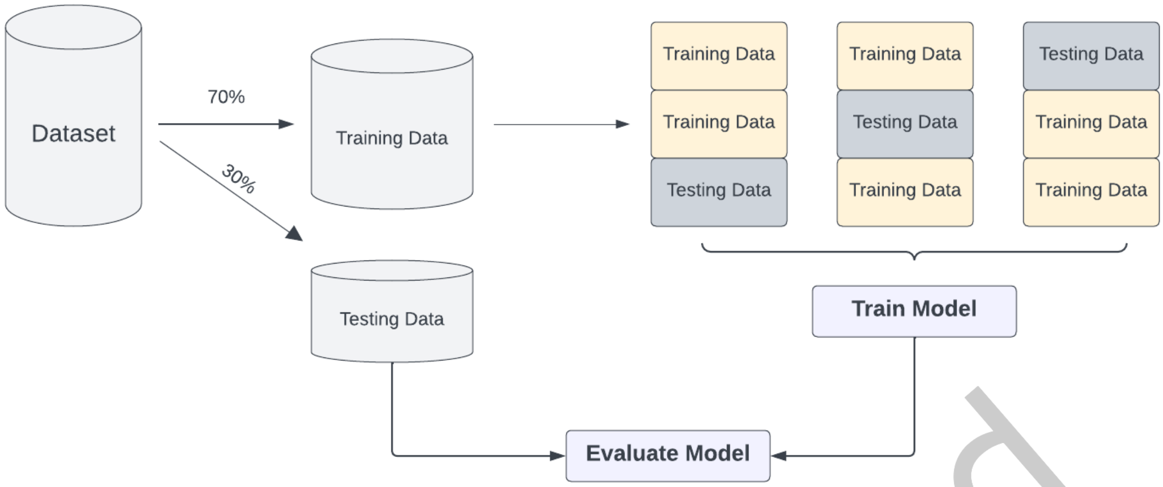

Fig. 10. K-fold cross-validation approach

Model	Embedding	Precision	Recall	F1-Score	Accuracy
Multimodal	VGG-16+BERT(Iteration-1)	0.83	0.82	0.82	0.82
	VGG-16+BERT(Iteration-2)	0.79	0.77	0.77	0.80
	VGG-16+BERT(Iteration-3)	0.80	0.84	0.81	0.80
	VGG-16+BERT(Testing Dataset)	0.79	0.81	0.79	0.81

Table 3. K-Fold Model Performance Comparison of imbalanced dataset

neural network that helps generate rich feature representations for a wide range of images. It contains 18 layers, whereas VGG16 contains 16 layers. The pre-trained models were fine-tuned for extracting the features from the model, and images were re-sized to make them compatible with the model. After that, the models were trained, and the features of the trained models were extracted. Our proposed method used a pre-trained version of ResNet-18 and VGG16 using the PyTorch framework.

4.3.2 Embedding Concatenation. After generating the embeddings or features for the text and image separately, these embeddings were concatenated in three different ways:

- Image embedding generated using ResNet-18 is concatenated with the text embeddings generated by using LASER and LaBSE, respectively.
- All three embeddings i.e. LASER, LaBSE, and ResNet-18, were concatenated.
- Concatenated the Image embedding generated using VGG16 and text embedding generated by Multilingual BERT.

4.3.3 Model Training. For training the model for detecting offensive memes, different classifiers were used. The logistic regression(LR) classifier gave the best result among all the classifiers. So, Logistic Regression was used as our classifier for training and calculating the performance of the three concatenated embeddings. All three concatenated embeddings (as mentioned in section 4.3.2) were fed as input along with the labels to the classifier separately, and their performances were compared.

Comparison	T-Statistic	P-Value	95% Confidence Interval	Degrees of Freedom
F1-Scores (Ensemble vs Unimodal, Imbalanced Data)	0.4288	0.6793	(-0.122, 0.195)	7
F1-Scores (Ensemble vs Unimodal, Balanced Data)	0.2729	0.7918	(-0.092, 0.125)	7

Table 4. Two-sample t-test results comparing F1-scores of ensemble and unimodal models for imbalanced and balanced data

5 Results and Discussion

The problem is a classification problem. Hence, the performance metrics of classification models, i.e., precision, recall, accuracy, and f1-score, are used to evaluate the models. A two-sample t-test was conducted to compare the performance between the ensemble models and other unimodal approaches. For the evaluation of all the models, 6946 memes were used for training, and 2316 memes were used for testing from the imbalanced dataset, whereas for the balanced dataset, 2799 memes were utilized for training, and 933 memes were allocated for testing. For k-fold, the training dataset was further divided into 3 parts and testing data was used for evaluating the final model(VGG-16+BERT). Model performance comparison is given in Table 6 and Table 7 for imbalanced and balanced datasets, respectively. Figure 11 and Figure 12 compare the text unimodal results for balanced and imbalanced datasets, respectively. Figure 13 and Figure 14 compare the performance of multimodal for balanced and imbalanced, respectively. Figure 15 shows the performance of VGG-16+BERT performance for k-fold validation.

5.1 Comparison

This subsection works upon the comparison between the results obtained in this research work and other research works on hateful meme detection. Table 5 shows the comparison between the accuracy or f1-score obtained in various research papers on hateful meme detection that uses the relation between image and text. The best accuracy for [21] Accuracy 0.83, for [24] is 0.74, for [42] is 0.71, for [31] is 0.73, for [4] is 0.75 and the best f1-score for [13] is 0.55 and [42] is 54.7 whereas the best accuracy and f1-score obtained in this research work is 0.83 and 0.82 which exceeds all the above-stated papers. We have presented final results for both imbalanced dataset and balanced dataset. It is to be noted here that our final dataset of 9262 meme images is imbalanced in nature. This imbalance towards offensive data points can be attributed to the fact that most social media platforms report the statistics that presence of hateful content is lower than that of non-hateful content on these platforms. [9].

5.2 Two-sample t-test

The two-sample t-test is a statistical test that compares the means of two different independent groups and determines whether they differ significantly from each other or not. The two-sample t-test between f1-scores of the ensemble models and other unimodal were calculated. The t-statistic value between the two was 0.4288 and a p-value of 0.6793 for the imbalanced data and 0.2729 t-statistic value and 0.7918 p-values for the balanced data. The positive value of the t-statistic suggests that the ensemble models perform better than the unimodal ones. Table 4 provides a comparison of these results, including the confidence intervals and degrees of freedom for a more comprehensive analysis.

5.3 Precision

Precision is defined as the accuracy of positive predictions. It checks the aptness of the classification model. Precision can be calculated using Equation (1). For the imbalanced dataset, the best precision of 0.82 was obtained from text unimodal(LASER+LaBSE and Multilingual BERT), whereas, for the balanced

Citation	Dataset	Best Result
[21]	7416 Hindi memes	Accuracy 0.83
[17]	7418 Bengali memes	Weighted F1 0.72
[24]	25000 Hinglish memes from Facebook, Twitter, Reddit	Accuracy 0.74
[42]	Combination of memes based on COVID-19 and US Politics	Macro F1 54.7
[13]	2300 total memes of Tamil, Malayalam and Kannada	Weighted avg F1 0.55
[31]	Memes dataset from Reddit and Gab	Accuracy 0.73
[4]	Facebook Hateful Meme Dataset	Accuracy 0.75

Table 5. Result Comparison

dataset, Multilingual BERT gave the best precision of 0.71.

$$Precision = \frac{TruePositive}{TruePositive + FalsePositive} \tag{1}$$

5.4 Recall

Recall refers to the ability of a model to detect positive predictions. It is the ratio of true positive to the total positives, i.e. true positive + false negative (Equation (2)). The higher the value of recall, the more will be the number of correctly predicted positive values. For the imbalanced dataset, Multilingual BERT gave the best recall of 0.83, and for the balanced dataset, Multilingual BERT and VGG-16 + BERT gave the best recall of 0.71, which means they predicted the highest true positives.

$$Recall = \frac{TruePositive}{TruePositive + FalseNegative} \tag{2}$$

5.5 F1-score

F1-score is the harmonic mean of recall and precision. The data used for this project is imbalanced, and the F1-score is especially beneficial for imbalanced data. F1-score can be calculated using Equation (3). For the imbalanced dataset, Multilingual BERT gave the best f1-score of 0.82, and for the balanced dataset, Multilingual BERT and VGG-16 + BERT gave the best recall of 0.70

$$F1 - score = \frac{2 * Precision * Recall}{Precision + Recall} \tag{3}$$

5.6 Accuracy

For the imbalanced dataset, CNN gave the best accuracy of 0.84, and for the balanced dataset, LASER+LaBSE+ResNet-18 gave the best accuracy of 0.80

$$Accuracy = \frac{TruePositive + TrueNegative}{TruePositive + FalsePositive + TrueNegative + FalseNegative} \tag{4}$$

5.7 Error Analysis

Despite attaining a significant accuracy, the obtained results are not perfect. This error can be attributed to the following factors:

Model	Embedding	Precision	Recall	F1-Score	Accuracy
Text Unimodal	LASER	0.78	0.81	0.76	0.81
	LaBSE	0.79	0.82	0.79	0.82
	LASER+LaBSE	0.82	0.82	0.77	0.82
	Multilingual BERT	0.82	0.83	0.82	0.83
Image Unimodal	CNN	0.59	0.39	0.47	0.84
	VGG-16	0.79	0.82	0.79	0.82
Multimodal	LASER+ResNet-18	0.72	0.73	0.72	0.77
	LaBSE+ResNet-18	0.79	0.73	0.75	0.79
	LASER+LaBSE+ResNet-18	0.79	0.80	0.79	0.80
	VGG-16+BERT(Testing Dataset)	0.79	0.81	0.79	0.81

Table 6. Model Performance Comparison of imbalanced datase

Model	Embedding	Precision	Recall	F1-Score	Accuracy
Text Unimodal	LASER	0.65	0.66	0.65	0.66
	LaBSE	0.67	0.69	0.67	0.68
	LASER+LaBSE	0.65	0.69	0.66	0.68
	Multilingual BERT	0.71	0.71	0.71	0.71
Image Unimodal	CNN	0.46	0.39	0.47	0.64
	VGG-16	0.69	0.68	0.68	0.68
Multimodal	LASER+ResNet-18	0.67	0.64	0.65	0.77
	LaBSE+ResNet-18	0.69	0.63	0.64	0.79
	LASER+LaBSE+ResNet-18	0.61	0.65	0.62	0.80
	VGG-16+BERT	0.70	0.71	0.70	0.70

Table 7. Model Performance Comparison of balanced dataset

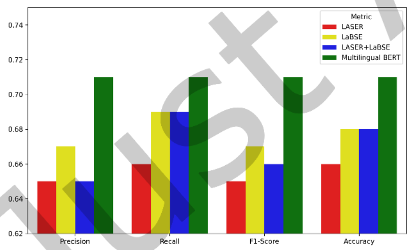

Fig. 11. Balanced Dataset- Unimodal performance w.r.t embedding

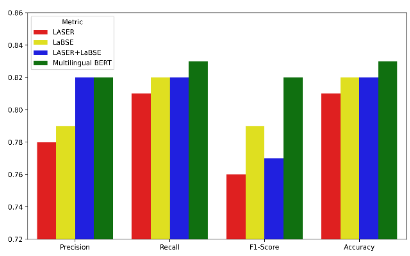

Fig. 12. Imalanced Dataset- Unimodal performance w.r.t embedding

- The memes are generally created based on the context of current events happening around the world. The models can't generally learn about each and every incident to predict if a given meme propagates hatred based on a specific one.
- Although we have tried to remove all the outliers, but we cannot deny the existence of outliers in the database.

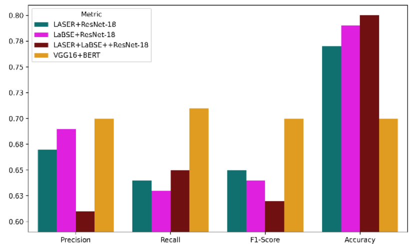

Fig. 13. Balanced Dataset- Multimodal Performance

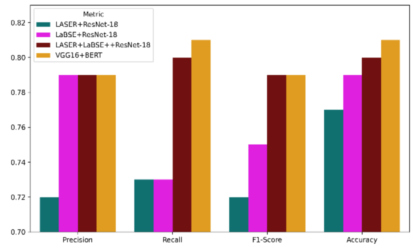

Fig. 14. Imbalanced Dataset- Multimodal performance

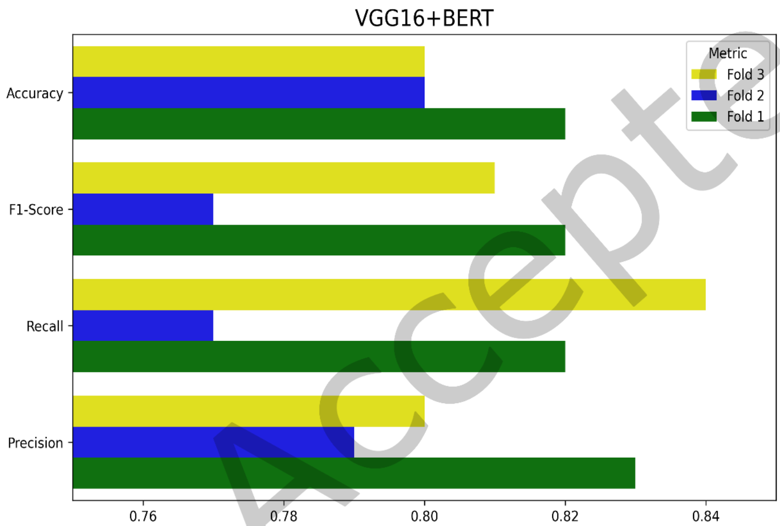

Fig. 15. VGG16+BERT Performance in 3 Folds

6 Conclusion and Future Scope

This paper presented a new dataset of Hindi memes and various models to classify Hindi memes as offensive and benign. Considering the f1 score of offensive memes, multimodal gave the best results and the logistic regression algorithm for text unimodal gave the best accuracy.

As discussed in the results section, this research work presents a multimodal approach for offensive Hindi memes detection and this multimodal approach exceeds the accuracy and f1-score obtained from the papers in Table 5.

In this paper, only two labels, benign and offensive memes, were defined. This can be increased to the degree of offensiveness. Not offensive, slightly offensive, medium offensive, or highly offensive can be an example of classification through the degree of offensiveness.

The models can be improved by using highly pre-trained models like Visual BERT, fine-tuned VL-BERT, etc.

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